

AULA BÔNUS 13

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- EXPRESSÕES NUMÉRICAS

Dso

DIREITO SIMPLES E OBJETIVO

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EXPRESSÕES NUMÉRICAS (parte 1)

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Ordem de realização das contas:

1. Parênteses ✓

()
2. Colchetes ✓

[]
3. Chaves ✓

{ }
- $-2 - 5 = -7$

Regra do sinal

+	÷	+	=	+
+	÷	-	=	-
-	÷	+	=	-
-	÷	-	=	+

Ordem de realização das operações:

1. Potenciação
2. Divisão e multiplicação
3. Soma e subtração



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CUIDADO!!!

Qual o valor da expressão aritmética ao lado $6 \div 2 \cdot 3$?

- A) 1
- ~~B) 9~~
- C) 6
- D) 1,2
- E) 3



$3 \cdot 3$

9

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EXERCÍCIO DE FIXAÇÃO

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1. Faça as seguintes contas (deixe o resultado o mais simplificado possível):

a) $\frac{3}{2} \cdot \frac{1}{4} - \frac{3}{8} + \frac{4}{10} \div \frac{8}{5} + \frac{1}{4} =$

$\frac{3}{8} - \frac{3}{8} + \frac{4}{10} \times \frac{5}{8} + \frac{1}{4}$

$\frac{1}{4} + \frac{1}{4}$

$\frac{2}{4} = \frac{1}{2} \checkmark$

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b) $\frac{3}{\cancel{8}_1} \cdot \left(\frac{-\cancel{2}^1}{8} \right) + \frac{1}{4} =$

$$-\frac{3}{8} + \frac{1}{4}$$

$$\frac{-3+2}{8} = -\frac{1}{8}$$

$$\begin{array}{r|l} 4-8 & 2 \\ 2-4 & 2 \\ 1-2 & 2 \\ \hline 1-1 & 2 \times 2 \times 2 \times 8 \end{array}$$

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c) $4 - \underline{6 \cdot 4} + \underline{45 \div 5} - 5 =$

$$\underline{4} - \underline{24} + \underline{9} - \underline{5}$$

$$13 - 29$$

$$-16$$

$$\begin{array}{r} 29 \\ -13 \\ \hline 16 \end{array}$$

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d) $5 - \left\{ 3 \cdot \frac{1}{3} \cdot \left[\frac{7}{(30 - 5 \cdot 5 + 3)} + 5 \right] \right\} - 10 =$

$$5 - \left\{ 3 \cdot \frac{1}{3} \cdot \left[\frac{7}{(30 - 25 + 3)} + 5 \right] \right\} - 10$$

$$5 - \left\{ 3 \cdot \frac{1}{3} \cdot \left[\frac{7}{8} + 5 \right] \right\} - 10$$

$$5 - \left\{ \cancel{3} \cdot \frac{1}{\cancel{3}} \cdot \left[\frac{47}{8} \right] \right\} - 10$$

$$5 - \frac{47}{8} - 10$$

$$\frac{40 - 47 - 80}{8}$$

$$\frac{-87}{8}$$

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e) $\frac{1}{0,2} + \frac{0,4}{2} = \frac{10}{2} + \frac{4}{20} = \frac{100 + 4}{20} = \frac{104}{20} = \frac{52}{10} = \frac{26}{5}$

10

$$f) \left(\underline{\underline{4}} - \frac{5}{\underline{\underline{2}}} \right) \cdot \left(\underline{\underline{3}} + \frac{1}{\underline{\underline{3}}} \right) =$$

$$\frac{\cancel{1}^1 \cdot \cancel{10}^5}{\cancel{2}^2 \cdot \cancel{3}^1} = 5 //$$

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$$g) \left[1 + \left(\underline{\underline{1}} + \frac{1}{\underline{\underline{2}}} \right) \right] - \left[2 - \left(\underline{\underline{1}} + \frac{1}{\underline{\underline{2}}} \right) \right] =$$

$$\left[\underline{\underline{1}} + \frac{3}{\underline{\underline{2}}} \right] - \left[\underline{\underline{2}} - \frac{3}{\underline{\underline{2}}} \right]$$

$$\frac{5}{2} - \frac{1}{2} = \frac{4}{2} = 2 //$$

12

$$h) \left(\frac{4}{3} + \frac{7}{15} \right) \cdot \left(\frac{7}{5} + \frac{4}{15} \right)$$

$$\left(\frac{20+7}{15} \right) \cdot \left(\frac{21+4}{15} \right)$$

$$\frac{27}{15} \cdot \frac{25}{15}$$

$$\frac{27}{9} = 3$$

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$$i) \frac{1 - \frac{1}{2}}{1 + \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}$$

$$\frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3} \cdot \frac{2}{2} = \frac{1}{3}$$

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$$j) \frac{5}{2} - \frac{1}{2} \cdot \frac{4}{2} + \frac{1 - \frac{1}{2}}{1 + \frac{1}{2}} \quad //_3$$

$$\frac{5}{2} - 1 + \frac{1}{3}$$

$$\frac{15 - 6 + 2}{6} = \frac{11}{6} //$$

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$$k) \underline{3^3} + \underline{2^3} - 3 \times 2 =$$

$$27 + 8 - 3 \times 2$$

$$27 + 8 - 6$$

$$29 //$$

16

$$l) \{[(8.4 + 3) \div 7 + (3 + 15 \div 5) \cdot 3] \cdot 2 - (19 - 7) \div 6\} \cdot 2 + 12 =$$

$$\{ \overset{35}{[(32 + 3) \div 7 + (3 + 3) \cdot 3]} \cdot 2 - \underline{(12) \div 6} \} \cdot 2 + 12$$

$$\{ \overset{5}{[35 \div 7]} + \overset{18}{6 \cdot 3} \} \cdot 2 - 2 \} \cdot 2 + 12$$

$$\{ \overset{46}{23 \cdot 2} - 2 \} \cdot 2 + 12$$

$$44 \cdot 2 + 12$$

$$88 + 12 = 100 //$$

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$$m) \{4 + 2 \cdot [32 - \frac{1}{4} \cdot (\frac{2}{3} - \frac{1}{8}) + 2] + 16\} + 1 =$$

$$\{4 + 2 \cdot [32 - \frac{1}{4} \cdot (\frac{16-3}{24}) + 2] + 16\} + 1$$

$$\{4 + 2 \cdot [32 - \frac{1 \cdot 13}{4 \cdot 24} + 2] + 16\} + 1$$

$$\{4 + 2 \cdot [32 - \frac{13}{96} + 2] + 16\} + 1$$

$$\{4 + \cancel{X} \cdot [\frac{3072 - 13 + 192}{\cancel{96} \cdot 48}] + 16\} + 1$$

$$\{ \underline{4} + \frac{3251}{48} + \underline{16} \} + 1$$

$$\{ 20 + \frac{3251}{48} \} + 1$$

$$\{ \frac{960 + 3251}{48} \} + 1$$

$$\frac{4211}{48} + 1 = \frac{4211 + 48}{48}$$

$$\frac{4259}{48}$$

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$$n) 3 \cdot \left\{ -1 + 12 \cdot \left[-13 + 4 \cdot \left(\textcircled{1} - \frac{1}{\textcircled{3}} \right) - 1 \right] - 1 \right\} =$$

$$3 \cdot \left\{ -1 + 12 \cdot \left[-13 + 4 \cdot \frac{2}{3} - 1 \right] - 1 \right\}$$

$$3 \cdot \left\{ -1 + 12 \cdot \left[\frac{-13}{1} + \frac{8}{3} - \frac{1}{1} \right] - 1 \right\}$$

$$3 \cdot \left\{ -1 + \cancel{12}^4 \cdot \left[\frac{-39 + 8 - 3}{3} \right] - 1 \right\}$$

$$3 \cdot \left\{ -1 + 4 \cdot (-34) - 1 \right\}$$

$$3 \cdot \left\{ -1 - 136 - 1 \right\}$$

$$3 \cdot \left\{ -138 \right\}$$

$$\underline{\underline{-414}}$$

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EXPRESSÕES NUMÉRICAS

(parte 2)

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$$o) \left[\left(\frac{1}{2} \cdot \frac{1}{3} \right) + \frac{4}{6} \right] =$$

$$\left[\frac{1}{6} + \frac{4}{6} \right]$$

$$\frac{5}{6}$$

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$$p) \left[\left(\textcircled{1} + \frac{1}{\textcircled{2}} \right)^2 - 2 \right] =$$

$$\left[\left(\frac{3}{2} \right)^2 - 2 \right]$$

$$\frac{3^2}{2^2} - 2$$

$$\frac{9}{\textcircled{4}} - \textcircled{2}$$

$$\frac{1}{4}$$

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q) $\frac{1}{5} + \left\{ \left[\frac{4}{9} \div \left(\frac{1}{2} \cdot \frac{2}{4} - \frac{1}{9} \right) \right] \right\} =$

$$\frac{1}{5} + \left\{ \left[\frac{4}{9} \div \left(\frac{1}{4} - \frac{1}{9} \right) \right] \right\}$$
$$\frac{1}{5} + \left\{ \left[\frac{4}{9} \div \left(\frac{9-4}{36} \right) \right] \right\}$$
$$\frac{1}{5} + \left\{ \left[\frac{4}{9} \div \frac{5}{36} \right] \right\}$$

$$\frac{1}{5} + \left\{ \frac{4}{9} \times \frac{36}{5} \right\}$$
$$\frac{1}{5} + \frac{16}{5}$$
$$\frac{17}{5}$$

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r) $\left(\frac{2}{3} \cdot \frac{3}{2} \right) \div \frac{2}{3} =$

$$\frac{2}{3} \div \frac{2}{3} \longrightarrow \frac{2}{3} \times \frac{3}{2} = \frac{1}{1} = 1 //$$

$$1$$

24

s) $\frac{(4 - \frac{4}{5})}{\frac{1}{3}} \div \frac{(9 + \frac{1}{3})}{\frac{1}{3}} =$

$$\frac{16}{5} \div \frac{28}{3}$$

$$\frac{\overset{4}{\cancel{16}}}{5} \times \frac{3}{\cancel{28}^7} = \frac{12}{35}$$

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2. A fração $\frac{4^{48} + 2^{97}}{8^{32}}$ é equivalente a:

- A) 2^{96}
- B) 2
- ☒ C) 3
- D) 6
- E) 8

$$\frac{(2^2)^{48} + 2^{97}}{(2^3)^{32}} = \frac{2^{96} + 2^{97}}{2^{96}} = \frac{2^{96}}{2^{96}} + \frac{2^{97}}{2^{96}}$$

$2^{96-96} = 2^0$ $2^{97-96} = 2^1$

$\downarrow$

$1 + 2^1$

$1 + 2 = 3$

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3. O valor aproximado de $\frac{16^{-0,75} + \sqrt[5]{0,00243}}{\frac{2}{3} + 4,333...}$ é:

- A) 0,045
- B) 0,125
- C) 0,315
- ~~D) 0,085~~
- E) 0,25

$$\frac{(2^4)^{-\frac{3}{4}} + \sqrt[5]{\frac{243^\circ}{100000}}}{\frac{2}{3} + \frac{13}{3}}$$
$$2^{-3} + \sqrt[5]{\frac{3}{10}}$$
$$\frac{15}{3}$$

$$\frac{\frac{1}{8} + \frac{3}{10}}{5}$$

$$\frac{5 + 12}{40}$$
$$\frac{17}{40} \times \frac{1}{5}$$
$$\frac{17 \times 5}{200 \times 5} = \frac{85}{1000}$$
$$\underline{\underline{0,085}}$$

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GABARITO DAS QUESTÕES

1.
- a) 1/2

b) -1/8

c) -16

d) -87/8
- e) 26/5

f) 5

g) 2

h) 3
- i) 1/3

j) 11/6

k) 29

l) 100
- m) 4259/48

n) -414

o) 5/6

p) 1/4
- q) 17/5

r) 1

s) 12/35
2. C
3. D

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